Commitment To Cooperation With Self-Negotiated Contracts

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Why Contracts?

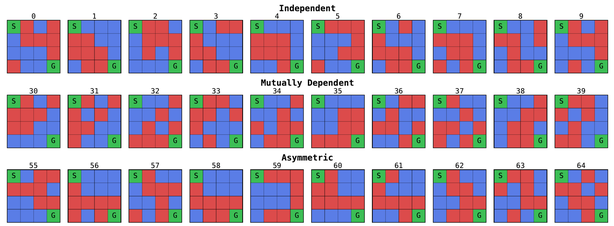
As AI agents operate autonomously in multi-agent settings, they must cooperate with other agents and with humans. Cooperation is hard because its **costs come early but its benefits come later**, creating an incentive to defect.

Human societies solve this principal–agent problem with **contracts**: observable representations of agreements that enable credible commitment through enforcement of terms.

Can LLM agents use self-negotiated contracts to cooperate with commitment?

Board Types (80 boards)

- **Independent (20):** each player can finish alone — tests rule-following without needing cooperation
- **Mutually Dependent (20):** neither player can finish without trading — tests finding mutual gains
- **Asymmetric (40):** P-Red can finish alone, P-Blue cannot — tests bargaining under power asymmetry



Metrics: Normalized Joint Reward, Gini (inequality), Both Finished, Both Beat Baseline (BBB), Defection.

Experimental Setup

Models	GPT-4.1, Haiku-4.5, LLaMA Maverick, LLaMA Scout, Qwen-3-235B, Qwen-3-30B
Protocol	Self-play on all 80 boards; metrics computed per board
Conditions	No-Contract baseline vs each of the three contract types

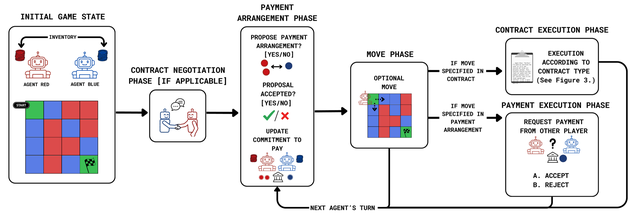
Without Contracts, Cooperation Fails Under Asymmetry

Pay-for-Partner, no contracts. R̂ = Normalized Joint Reward; BF = Both Finished; BBB = Both Beat Baseline; Def. = Defection.

Model	R̂	Gini	BF	BBB	Def.
GPT-4.1	0.72	0.28	0.43	**0.00**	0.23
Haiku-4.5	0.91	0.10	0.82	**0.00**	**0.01**
LLaMA Maverick	**0.97**	0.05	**0.95**	**0.00**	0.13
LLaMA Scout	0.84	0.15	0.71	**0.00**	0.53
Qwen-3-235B	0.66	0.23	0.44	**0.00**	0.39
Qwen-3-30B	**0.52**	0.26	**0.26**	**0.00**	**0.94**
Mean	**0.77**	**0.18**	**0.60**	**0.00**	**0.32**

BBB = 0.00 for *every* model: without contracts, no model ever got both players above their baseline on any Asymmetric board.

The CT-Bench Environment



Game	2 players (P-Red, P-Blue) on a 4×4 board; move to a tile by paying a chip of its color
Chips	Each player holds 14 chips of only its own color (+2 green), so trades are needed to cross the other color
Scoring	20 points for reaching the goal + 5 points per remaining chip; each player maximizes its own score
Pay-for-Partner	Players promise to cover each other's moves; promises are **non-binding**, so partners can renege

Self-Negotiated Contracts



Agents negotiate their own contracts, in three forms:

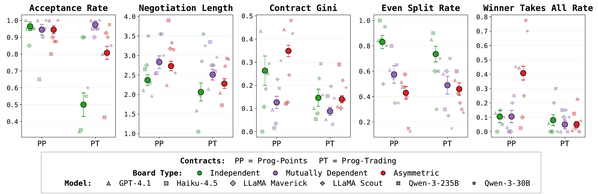
- **Programmatic Trading:** chip transfers to cover agreed tiles, auto-executed
- **Natural Language Trading:** contract in natural language, an LLM judge enforces it
- **Programmatic Points:** promise part of your final score, auto-paid

Negotiation runs up to 8 turns; a judge model (Qwen-3-235B) summarizes agreed terms into a JSON contract that both players must accept. The judge is highly accurate (0.26% error across 1,155 approved moves).

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{"(row,col)": {"giver": "Player X",
  "receiver": "Player Y", "color": "<Color>"}, ...}
```

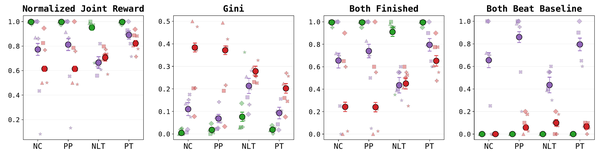
A Programmatic Trading contract: when the receiver steps on the tile, the chip transfer executes automatically; a giver who cannot pay scores 0.

Contract Formation



- Agents reliably reach agreement: **86%** mean acceptance (Haiku-4.5 an outlier at 67%, p<0.001)
- **Representation shapes fairness:** Programmatic Points yields winner-takes-all deals on 39% of Asymmetric boards (74% for Qwen models); Programmatic Trading assigns equal tile counts in 49% of cases

Contracts Improve Cooperative Outcomes

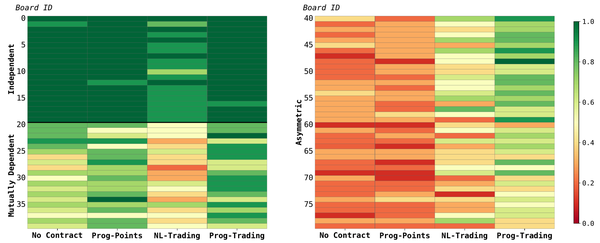


0.77 → 0.89
Normalized Joint Reward, No-Contract → Programmatic Trading (p<0.001)

–38%
fewer non-binding Pay-for-Partner arrangements under Programmatic Trading

45 vs 6
of 60 dependent boards where Programmatic Trading (vs No-Contract) has the highest Both Finished

Both Finished, Board by Board



Mean Both Finished across all 80 boards and contract types. Programmatic Trading gives small but consistent gains on Mutually Dependent and Asymmetric boards.

Contracts Help the Weaker Player

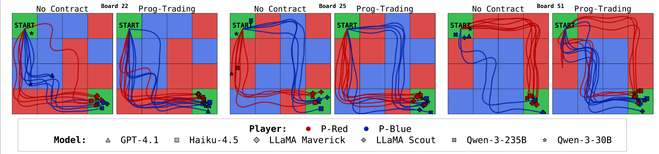
On Asymmetric boards, contracts primarily benefit the weaker player: under Programmatic Trading, P-Blue's average reward is **68% higher** than without contracts (p<0.001), while P-Red shows no significant gain — the stronger player fails to exploit its bargaining position.

When Contracts Backfire

On Mutually Dependent boards, natural-language contracts **underperform having no contract at all**: Both Finished is 46% vs 61% (No-Contract) vs 79% (Programmatic Trading).

Mechanism — over-anchoring: contracts formed are similar, but Natural Language Trading suppresses the follow-up trading still needed to finish (1.4 vs 3.0 proposals per game). When declining to trade, models cite their contract 94% of the time, vs 74% under Programmatic Trading.

Contracts Change Behavior on the Board



Contracts can improve outcomes. On boards where models fail to reach their goals without a contract, adding a Programmatic Trading contract lets every model complete its route.

Beyond Self-Play

Pairing GPT-4.1 with Haiku-4.5 (strong–strong) and Qwen-3-30B (strong–weak) on Mutually Dependent boards: under Programmatic Trading, the strong model lifts the weak model at its own expense while joint reward stays high — **efficient cooperation survives large capability gaps**.

Without contracts, GPT-4.1 scores 0 as P-Red against Qwen-3-30B, which honors only **6%** of its promises (Haiku-4.5 honors 100%). Programmatic Trading enforces the transfers and resolves this.

Takeaways

1. **The way contracts are represented and enforced matters:** contracts that compile to programs consistently outperform natural-language ones
2. **Self-negotiated contracts can improve cooperative outcomes** beyond what regular trading achieves
3. **Agents struggle to achieve mutually beneficial outcomes from asymmetric starting points** — structured commitment closes part of this gap

Paper & Acknowledgments

CT-Bench is open-sourced. Paper: **arxiv.org/abs/2607.22750**

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